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UNIT - IIIrd

ISOLATION, IDENTIFICATION AND ANALYSIS OF PHYTOCONSTITUENTS

① Syllabus →

Isolation, Identification and Analysis of Phytoconstituents

- Terpenoids: Menthol, Citral, Artemisinin
- Glycosides: Glycyrrhetic acid, Rutin
- Alkaloids: Atropin, Quinine, Reserpine, Caffeine
- Resins: Podophylotoxin, Curcumin.

① Classes → ② drugs

• Phytoconstituents →

These are chemical compounds that occur naturally in plants and thus compounds are medicinally active compound which are dissolved out from the cells and tissues from the plant by the process of extraction.

• Isolation →

It is the process of separation of single compound (desired) from the mixture of compounds present in extract (phytoconstituents).

• Identification →

It is the process in which we have to identified that isolated compound is correct or not or confirm their identity.

So, many identification tests are used for this purpose.

Tests are different for different classes.

• Analysis →

It is the process of determination of concentration of extract.

used different technique for this such as spectroscopy and chromatography
TLC, HPLC etc

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A. TERPENOIDS

- Terpenoids are hydrocarbon of plant origin (naturally) of the general formula (C_5H_8).
- They are available in volatile oil.
- eg. Menthol, Citral, Artemisinin etc..

Properties -

- Colourless
- They are soluble in organic solvents (alcohol, chloroform, etc.) but are insoluble in water.
- Most of the terpenoids are optically active.
- They easily get oxidized by oxidizing agents.
- They undergo polymerization and dehydrogenation in the ring.
- Also known as isoprenoids and also it contain Isoprene unit



Classification -

S.No.	Name	No. of Isoprene unit	Molecular formula
01.	Monoterpene or Terpene	2	$C_{10}H_{16}$
02.	Sesquiterpenoids	3	$C_{15}H_{22}$
03.	Diterpenoids	4	$C_{20}H_{32}$
04.	Triterpenoids	6	$C_{30}H_{48}$
05.	Tetraterpenoids or Carotenoids	8	$C_{40}H_{64}$

Uses -

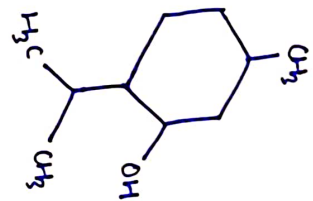
- Analgesic, Carminative, Anthelmintic, Antisepic etc..
- Also used in the preparation of Soap, Cosmetics, Perfumery + food articles.
- also used in pesticide and insecticide.

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1) Menthol -

It is belong to the class of monoterpene alcohol.



• Biological source -

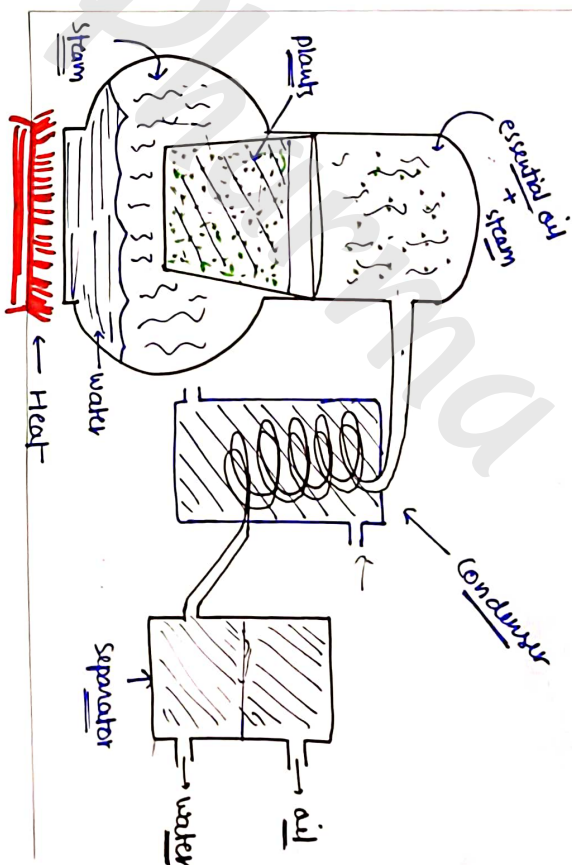
obtained from oils of mentha piperita, (black peppermint), white peppermint, M. piperita var. officinalis etc - family Labiatae.

• Isolation -

Menthol is isolated by two method i.e. Hydro-distillation and Steam distillation.

- Steam distillation →

- Take fresh air dried mentha plants and allow it into steam distillation
- The mixture vapour is generated (water and volatile oil) which further pass



through condenser.

- Now both vapour is cooled and back to liquid form (mentha oil + water)
- Mentha oil is floated on the top of water due to lighter than water which further separated out through separator.
- Now, mentha oil is filtered and allow it to cool; crystal of menthol will separate out.
↓
again re-crystallized for pure form

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• Identification →

few drop of sample
+

5ml of nitric acid

↓
Heated (on water bath)

Blue color obtained (within 5 minutes)

↓
after some time

Yellow color

↓ which indicate the presence of
methyl.

• Analysis (estimation) →

By TLC method (Thin layer chromatography)

• Sample preparation → 1 mg of methyl is dissolved in 1ml of methanol.

• Standard sample → Methanol

• Stationary phase → Silica gel-G

• Mobile phase → pure chloroform

• Detecting agent → 1% vanillin-sulphuric acid

reagent and heat the plate at
110°C for 10 minutes.

• Rf value → $\boxed{0.48 - 0.62}$ — standard.

procedure →

• Apply the spot of test and standard sample on silica gel G-plate.

• then elute it in pure chloroform solⁿ.

• Now, spray the dried plate with 1% vanillin-sulphuric acid reagent and heat the plate at 110°C for 10 min.

• Now, check the Rf value for test solⁿ and compare with standard.

• Utilization →

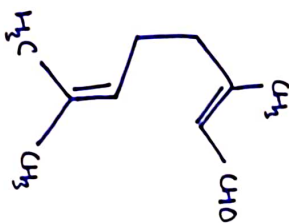
used for its cooling sensation, flavouring property, carminative etc--

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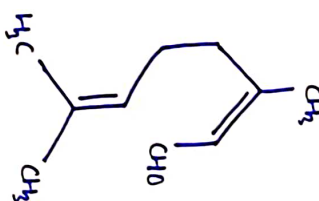
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2) Citral -

It is belongs to the class of monoterpen aldehydes.



Citral-a
(geranial)



Citral-b
(neral)

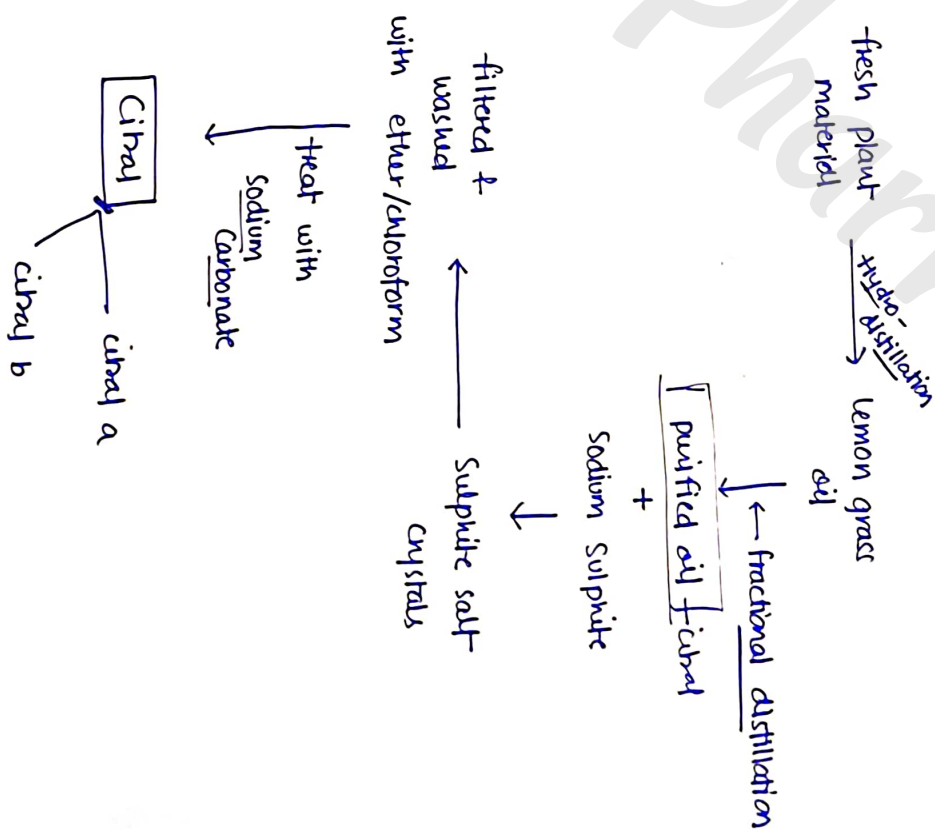
• Biological source →

Obtained from oils of lemon grass (75-85%)
flexuosus, C. martinii belongs to family Citraminae.

- It also present in lemon oil (Citrus limonius) and orange oil (Citrus aurantium).

• Isolation -

It is isolated by following method
i.e. hydro-distillation and fractional crystallization method.



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• Identification →

① Sample + alc. soln of

Sudan red III

Red colour

↳ which indicate presence of cinral.

②

Sample + Tincture alkaline

Red color

↳ which indicate presence of cinral.

• Analysis →

By TLC method

• Sample preparation → 1 mg of cinral is dissolved 1 ml of methanol.

• Standard sample → Cinral

• Stationary phase → Silica gel-G

• Mobile phase → Pure Chloroform

• Detecting agent → 2,4, dinitrophenyl hydrazine reagent

• Color spots → Yellow to orange

• Rf Value → 0.51.

Procedure →

• Same as methanol.

• Utilization →

- Used for perfumery and also as a flavouring agent.

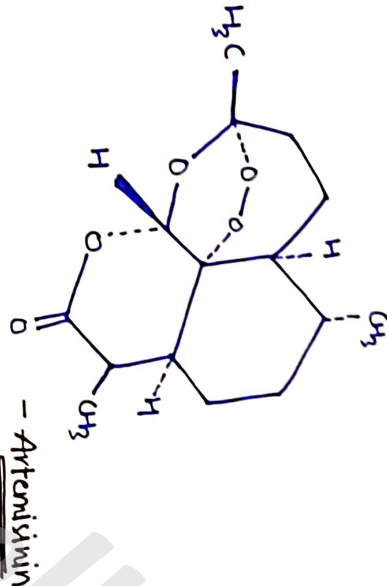
- It act as precursor for the synthesis of β -ionone which further used for synthesis of vit. A.

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3) Artemisin → Artemisinin

It is belongs to the class of sesquiterpene lactone.



• It is highly effective antimalarial drugs.

• Biological source →

Obtained from traditional Chinese medicinal plant herb Artemisia annua (sweet wormwood)

• Isolation →

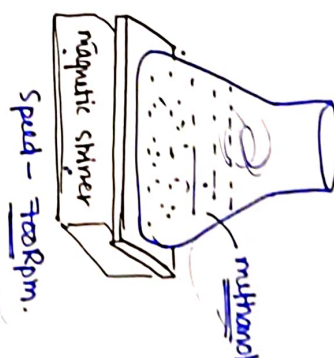
leaves (upper part of plant) → dried at air/sun

finer dried powdered

↓ macerated

Repeated until
methanol layer become colorless

↓



The extract is evaporator

by rota-evaporator at a

temp. of 40°C until volume

is reduced to 100ml

→ The methanolic extract is

partitioned using

50 ml hexane until

colorless hexane layer is obtained

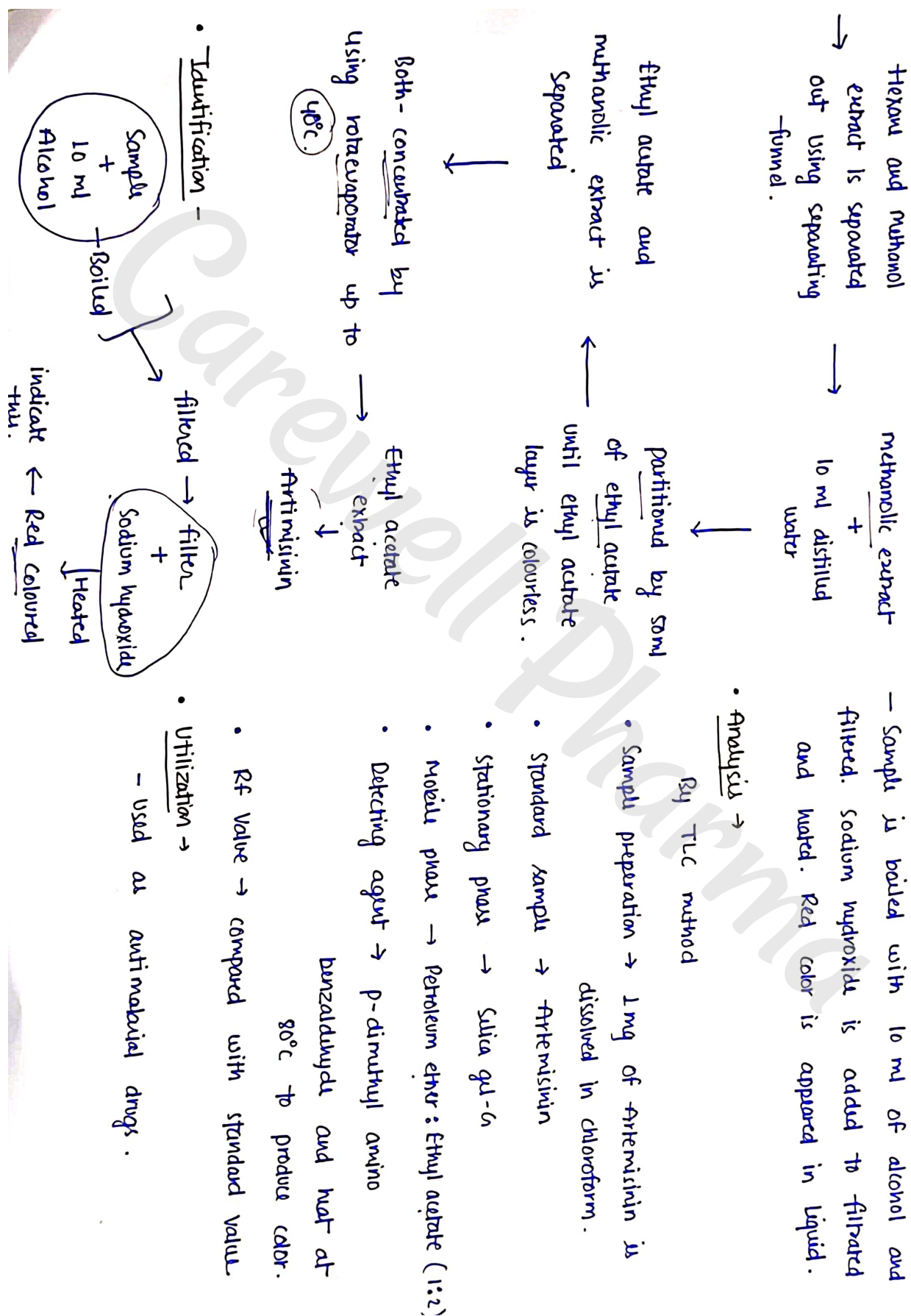
partitioned →



Hexane: Methanol (1:2)

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UNIT-3

Isolation, Identification and Analysis of Phytoconstituents

Syllabus →

- A) Terpenoids → Menthol, Citral, Artemisin
 B) Glycosides → Glycyrrhetic acid, Rutin
 C) Alkaloids → Atropin, Quinine, Reserpine, Caffeine
 D) Resins → Podophyllotoxins, Curcumin.

Important Questions →

Q. Write isolation, identification and analysis of →

- ✓ Menthol } terpenoids
- ✓ Citral }
- Caffeine } Alkaloids
- Reserpine }
- Atropin }
- Quinine }
- Curcumin } Resins
- Rutin } Glycosides

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B. GLYCOSIDES

It is an organic compounds obtained from plants and animals sources, which on enzymatic or acidic hydrolysis yields one or more sugar moieties (glycone) and a non-sugar moiety (aglycone).

properties →

- These are colorless but some of them are colored (eg. flavanoids - yellow, Anthracene - red).
- Crystalline or amorphous substance.
- Soluble in water + Alcohols, but insoluble in chloroform and ether.
- They are easily hydrolyzed by mineral acids, water and enzymes.

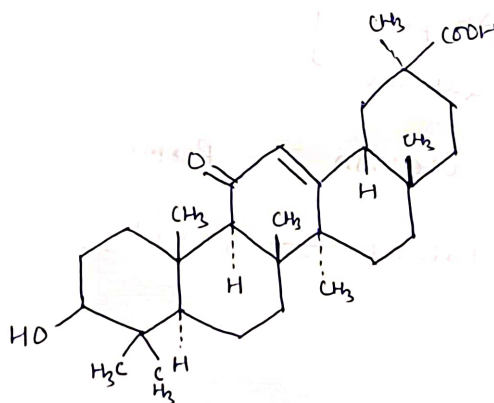
Drugs →

- ✓ i) Glycyrrhetic acid
- ✓ ii) Rutin

i) Glycyrrhetic acid →

• Synonyms →

Mulethi,
Liquorice root



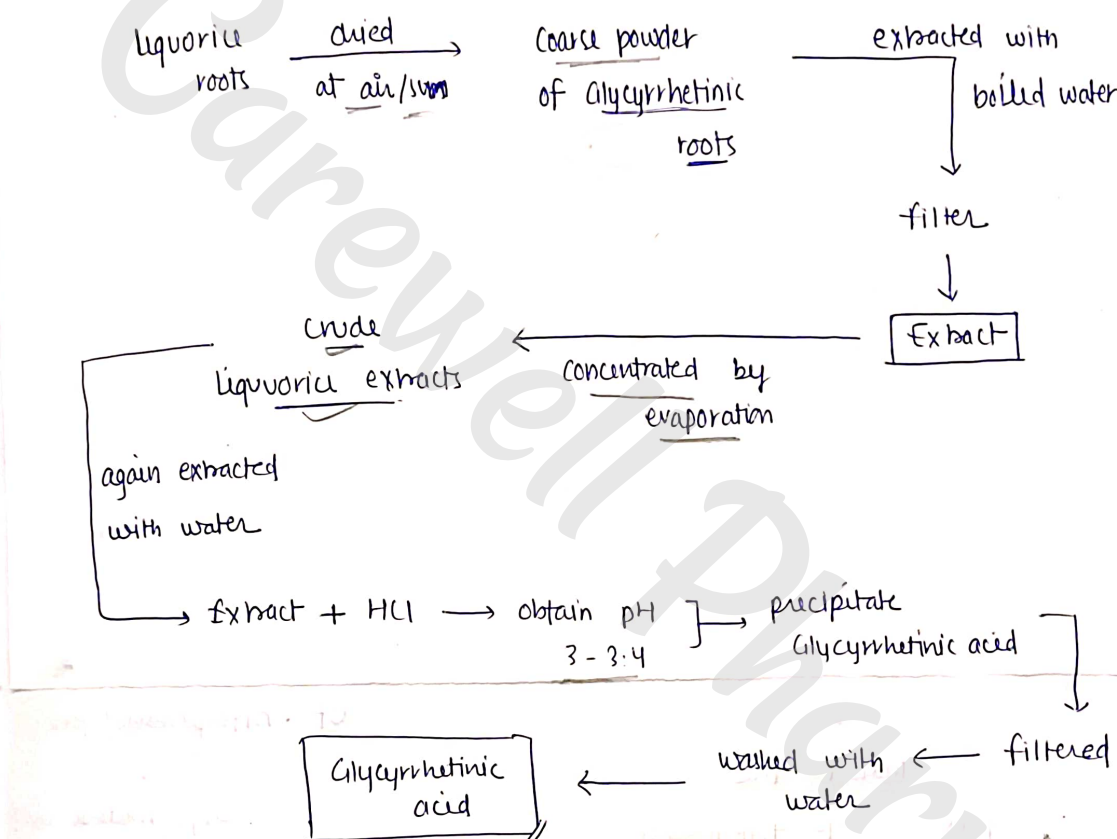
- Biological source → It is a triterpenoid saponin glycosides which

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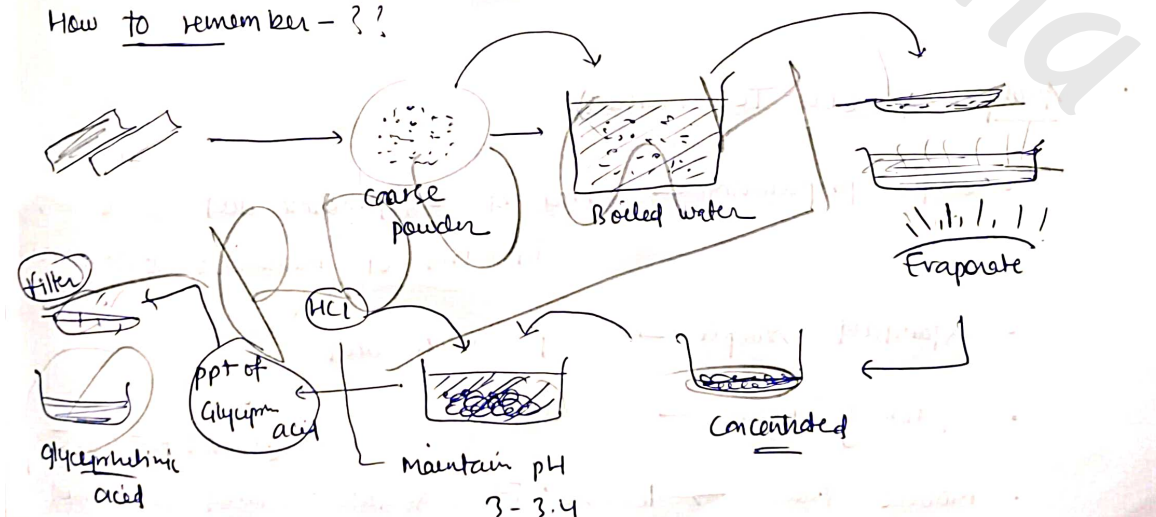
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obtained from the roots and stolons of Glycyrrhiza glabra
belongs to family - leguminosae.

Isolation →



How to remember - ??



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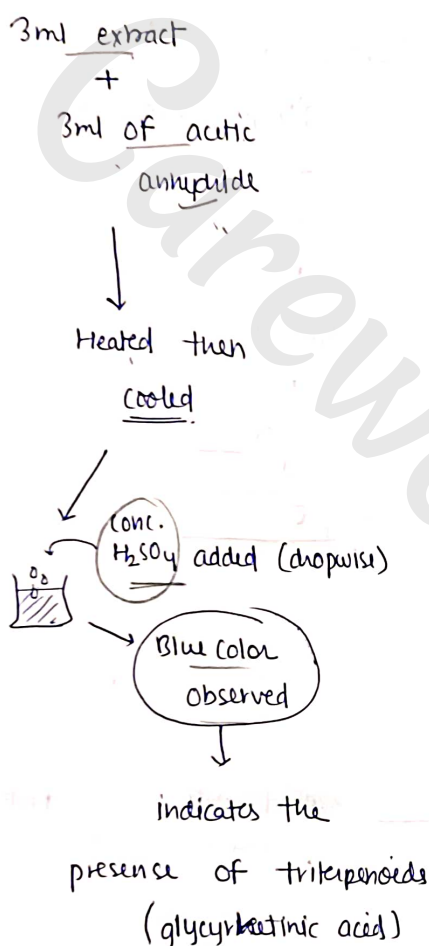
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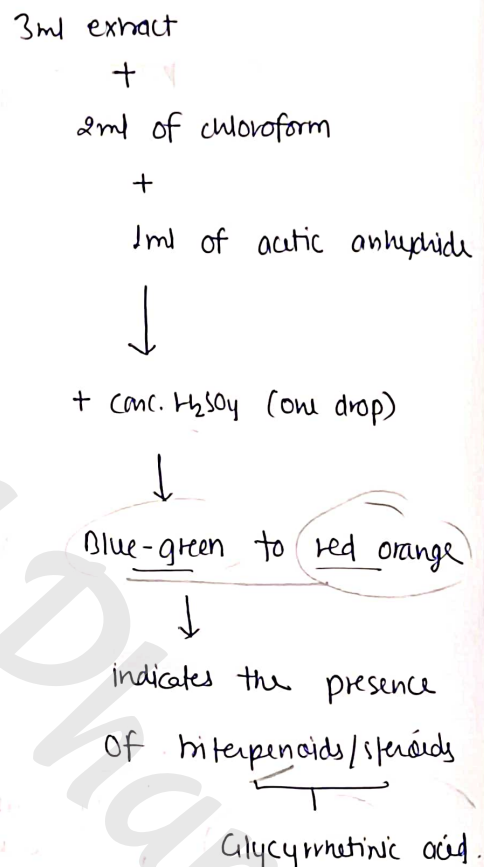
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• Identification →

• Liebermann test



• Liebermann-Burchard test



• Analysis → (By TLC method)

- Sample preparation → 1mg of Glycyrrhetic acid is dissolved in 1ml of methanol: chloroform (1:1).
- Standard sample → Glycyrrhetic acid
- Stationary phase → Silica gel-G (12.5 : 7.5 : 0.5)
- Mobile phase → Toluene : Ethyl acetate : Glacial acetic acid

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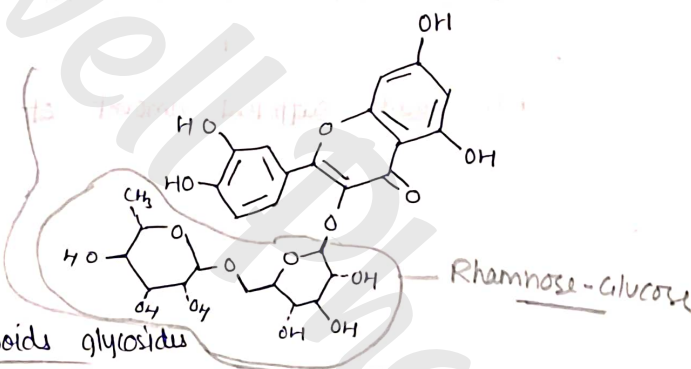
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- Detecting agent → 1% vanillin-sulphuric acid reagent
heated for 10 minutes at 110°C.
- Rf value → 0.41
- Color spot → Purplish spot
- Utilization (use) →
 - It is used to treat Rheumatoid Arthritis
 - Anti-inflammatory action
 - Used in Addisons disease

ii) Rutin →

- Synonym → Rutin
- Biological source →



- It is a flavonoids glycosides
- Obtained from the powdered of dried food grains of fagopyrum esculentum belongs to family Polygonacea.
- Also obtained from various citrus fruits.

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Isolation →

Take weight quantity of fagopyrum powder [cibus fruits powder]
then extract it with 80% ethanol in soxhlet
apparatus for 12 hrs.



filter the mixture and add 25 ml of water to ~~ex~~ filtrate



Now, treat this mixture with petroleum ether
then separate the petroleum ether through
separating funnel



Now, again aqueous layer treated with chloroform
and separate the chloroform layer



Now, cool aqueous layer then it will form
yellow precipitate



Now, separate this yellow ~~filtrate~~ with precipitate and
dissolved it in hot methanol



filter it and evaporate



Yellow powder of rutin will
be obtained.

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• Identification →

• Shenoda test

3ml ethanol
+
Sample
+
few drops of
 H_2SO_4



mixture
+
0.5g of
magnesium
turnings



pink color

• Sample
+
Lead acetate



Yellow color
precipitated

• Sample
+
ferri chloride



Dark green color

• Utilization →

- used as an antioxidant.
- Useful in the treatment of retinal haemorrhages.

• Analysis → (By using TLC method)

- Sample preparation → 1mg of Rutin is dissolved in 1ml of ethanol.
- Standard sample → Rutin.
- Stationary phase → Silica gel-G
- Mobile phase → Ethyl acetate : butanone : formic acid : water
solvent phase
(50:30:10:10) or (100:10:11:27)
- Detecting agents → Anisaldehyde sulphuric acid reagent
- Rf value → 0.43 with yellow color spot.

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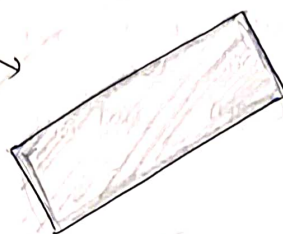
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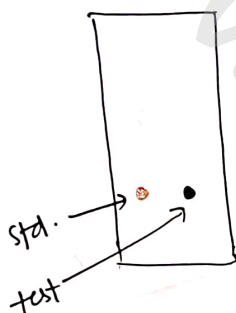
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Demo of Analysis by TLC

1. prepare silica gel G-plate



make two spot
from test + std sample



elute
with
mobile
phase



spot moves ~~for~~ upward
but invisible

distance
travel
calculated
in the
form of
Rf. value



spot will
be visible

apply detecting
agent

compare this
value for
std + test

& also check color of spot,

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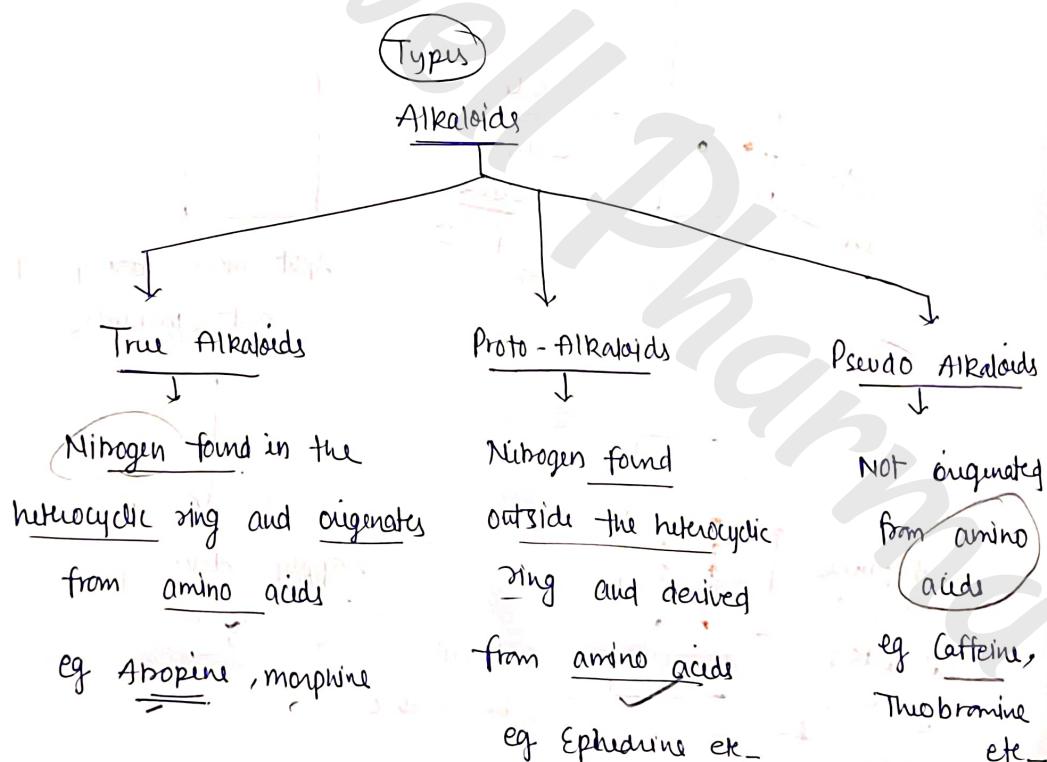
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Imp. C. ALKALOIDS

- These are basic nitrogenous compounds of plants or animals origin and possesses a pharmacological action.
- They contain nitrogen and mainly in heterocyclic ring
- mainly derived from amino acids
- founds in animals, fungi & bacteria.



properties →

- colourless, crystalline, non-volatile, solids (some are liquid & volatiles).
- Generally bitter in taste.

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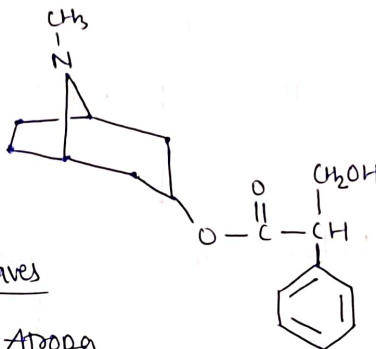
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Drugs → Atropine, Quinine, Reserpine, Caffeine

i) Atropine →

It is a tropane alkaloids obtained from the fresh or dried leaves and flowering tops of Atropa belladonna, family → Solanaceae



• It is insoluble in ether, easily soluble in chloroform, alcohol and sparingly soluble in water.

• Isolation →

Powdered drug material is taken and moistens with sodium carbonate solⁿ

↓

mixture is extracted in petroleum ether then filter it

↓

In filtrate, add aqueous acetic acid then extract it with ether

↓

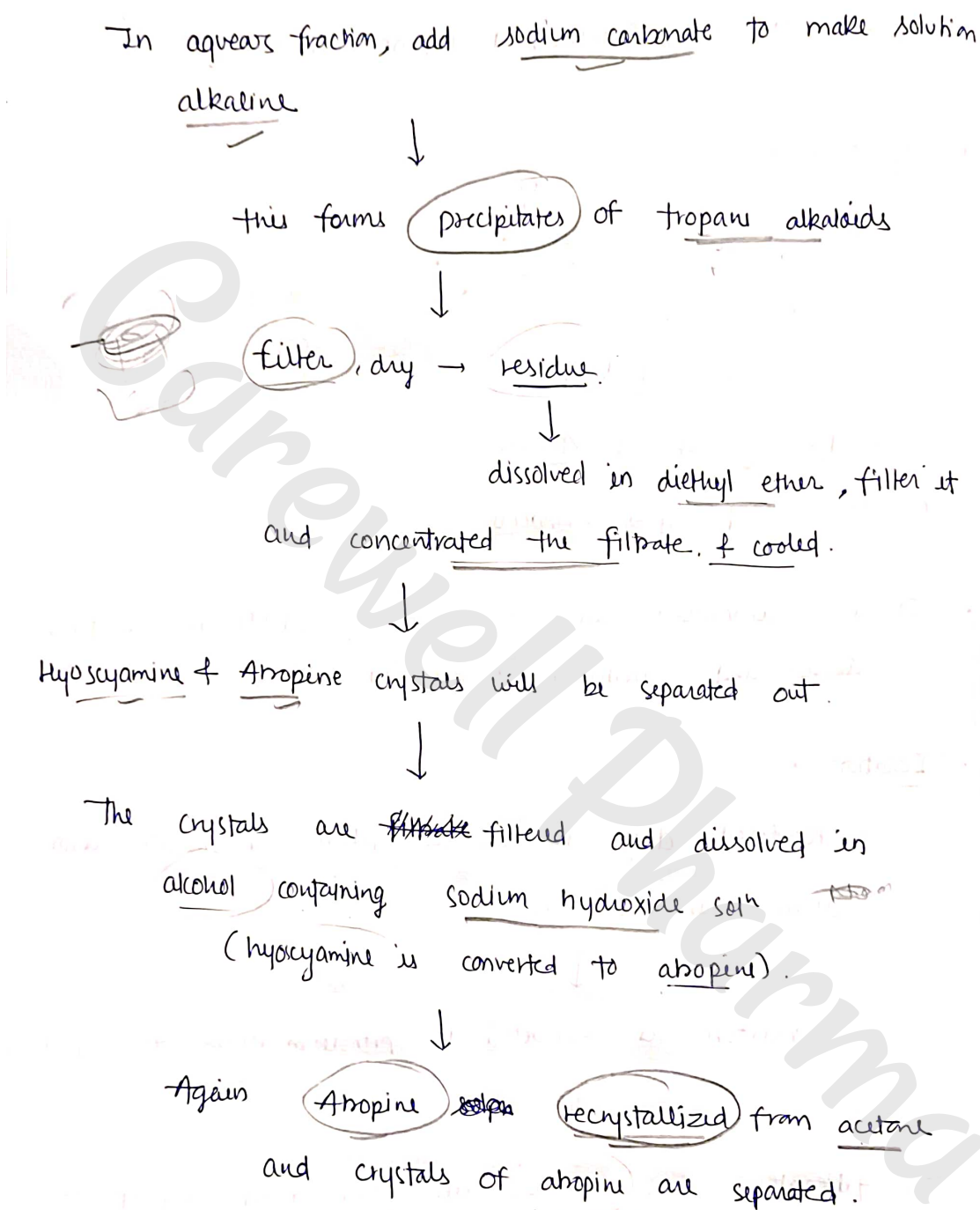
Both are separated through separating funnel. and

separate aqueous fraction.

↓

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• Identification →

• vitali-morin test →

Small quantity of the solid atropine is taken and added 2 drops of conc nitric acid

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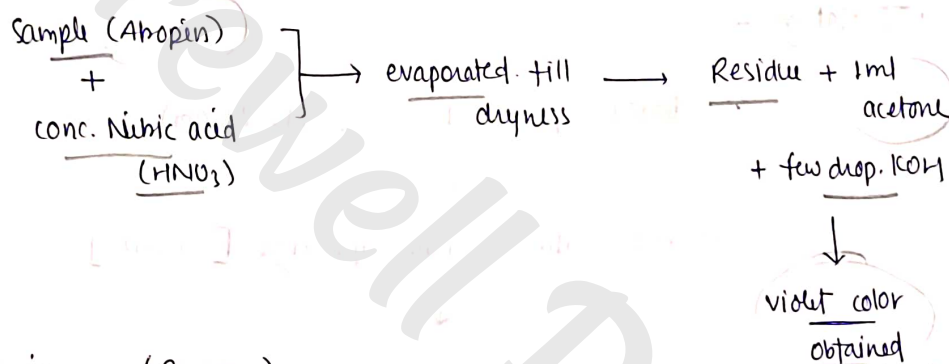
in an evaporating dish and evaporated to dryness on water bath.

↓

Then residue is dissolved in 1ml of acetone and few drops of freshly prepared alc. KOH is added

↓

Violent color residue is obtained.



• Analysis → (By TLC)

- Sample preparation → 1mg of Atropine dissolved in 1ml of chloroform.
- Standard sample → Atropine
- Stationary phase → Pre-coated silica gel on plate
- Mobile phase → Toluene : Ethyl acetate : Diethyl amine (70:20:10)
- Detecting agent → Dragendorff's reagent
- Rf value → 0.70
- Color spot → Yellow-orange spot
- Uses → Antispasmodic, antidote in opium poisoning

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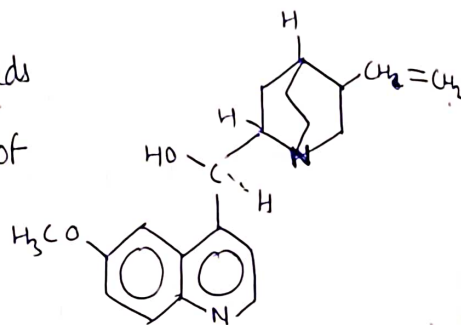
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ii) Quinine →

It is an quinoline alkaloids
 obtained from the dried bark of
Cinchona officinalis
 and belongs to family Rubiaceae



• Isolation -

Dried powdered bark (15-20 gm)



Mixed with calcium hydroxide $[Ca(OH)_2]$



Add 5% NaOH solⁿ to make paste



Allow to stand for few hours



extraction with Benzene by soxhlet continuous
 extraction process (about for 6hrs).



Shake the above extract with successive
 portion of 5% H₂SO₄ (sulphuric acid) in separating
 funnel.



Now, the aqueous acid extract is separated



Adjust the pH 6.5 with dil. NaOH



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cooled



Crystals of quinine sulphate are formed, filtered
and recrystallized with hot water.

• Identification →

Sample
+
Ammonium
solution
+
Bromine
water

→ emerald green indicates the presence of quinine

• Analysis →

- Sample preparation → 1mg of Quinine is dissolved in 1ml of methanol
- Standary sample → quinine
- Stationary phase → Silica gel - G₆₀
- Mobile phase → chloroform : Diethyl amine (9:1)
- Detecting agent → Dragendorff's reagent
- Rf value → 0.17

• Uses →

- It is an antimalarial drugs used to treat malaria.
- It also have cardiac depressant activity used to treat arrhythmias.

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iii) Reserpine →

It is an indole alkaloids obtained from the dried roots of Atropa belladonna, Rauwolfia serpentina, belongs to family - Apocynaceae.

• Isolation -

Rauwolfia root powder is exhaustively extracted with 90% alcohol in soxhlet apparatus.



Alcoholic extract is filtered, concentrated and dry dried under reduced pressure below 60°C to yield dry extract. evaporate



Again this dry extract is extracted with Ether : chloroform : Alcohol [20:8:2.5] then filtered.



In filtrate, dil. ammonia is added with intermittent shaking. कल-कल कर



Then water is added to precipitate the crude alkaloidal mixture and allow the drug to settle after vigorous shaking.



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↓
The solution is filtered off and extracted the residue with
4 volumes of 0.5N ammonium sulphate in separating

↓
The extract is made alkaline with dil. ammonia to liberate
alkaloid.

↓
finally it is extracted with 3 portions of chloroform.

↓
chloroform extract is collected, concentrated and evaporated
on water bath to yield rauwolfia alkaloids.

↓
Reserpine are separated by chromatography.

• Identification -

sample
+
solⁿ of vanillin
in acetic acid

} a violet red
colour is produced

Uses →

- antihypertensive
- antipsychotic agents.

• Analysis -

• sample preparation → 1 mg of reserpine is dissolved in
1 ml of methanol.

• Standard solⁿ → Reserpine

• Stationary phase → Silica gel-G

• Mobile phase → chloroform : acetone : Diethyl ether (10:40:10)

• Detecting agent → Dragendorff's reagent

• Rf Value → 0.72 with Orange spot.

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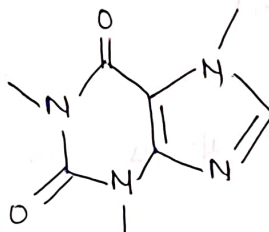
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iv) Caffeine →

It is a purine alkaloids

Obtained from Tea leaves, coffee seeds, cocoa,
and other species.



• Isolation -

①

- powder tea leaves is
extracted with boiling water

then aqueous extract is filtered



The warm extract is treated with lead acetate
to precipitate tannins and filtered



The filtrate is treated with excess of dil.
sulphuric acid to precipitate lead in the form of
lead sulfate.



filtered solⁿ is boiled with ^{activated} charcoal to remove
coloring if any, then remove charcoal through filtration.



filtered decolourised solⁿ is extracted with
chloroform



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↓
Combined the chloroform extracts and evaporated on
water bath to yield caffeine (white ppt).

↓
Recrystallized with alcohol.

②

powdered tea leaves are extracted with
ethanol in soxhlet extraction.

↓
Now, this ethanol contain Caffein which
further adsorbed on magnesium oxide (MgO).

↓
then caffeine is disorbed after treatment with
10% H₂SO₄

↓
Then it is extracted with chloroform and
recrystallized.

• Identification →

• Murexide test →

Sample
+
HCl
+
Potassium chlorate

heated to
dryness

show purple
color vapours
of dil. NH₄
ammonia

purple color is lost
on addition of
fixed alkali

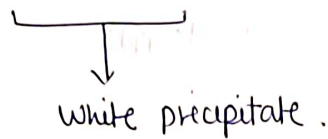
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- Caffeine + tannic acid



Analysis - (By TLC)

- Sample preparation → 1 mg of Caffeine is dissolved in 1 ml of methanol/chloroform
- Standard ~~soln~~ sample → Caffeine → silica gel - G₁
- Stationary phase → Ethyl acetate : methanol : acetic acid
mobile phase (80:10:10)
- Detecting agents → Vapours of iodine (expose this into plate)
- Rf value → 0.41 with color spot → Brown spot.

(By HPLC)

- method → Isoctetic
- Stationary phase → C₁₈ column
- Mobile phase → methanol : water (25:75)
- Detection → UV-Visible detection 254nm

Uses →

- Used as a CNS ↑ & Diuretics.

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D. RESINS

They are solid/semi-solid, amorphous products derived from natural living sources and mostly from the plant origin.

- they are mixtures of essential oils, oxygenated products of terpene and carboxylic acids.
- ~~They~~ found as exudations from the trunk of various trees.

Properties -

- These are translucent or transparent solid, semisolid, amorphous or sometimes liquids. [carbon compounds].
- Most of the resins are heavier than water.
- these are water insoluble but mostly soluble in alcohol, volatile oil, fixed oil and also some non-polar organic solvents like benzene or ether.

Drugs → Podophyllotoxin, Curcumin.

i) Podophyllotoxin →

It is the lactone resin present in the root and rhizome of Podophyllum hexandrum and P. emodi, belonging to family - Berberidaceae.

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• Isolation -

Roots of P. emodi is ^{extracted} ~~taken~~ with methanol in
soxhlet apparatus
 ↓
 filtered it and evaporate to semisolid mass
 ↓
 this semisolid mass is dissolved into acidic water
 (H₂O + little amount of HCl)
 ↓
precipitate is formed which should be allowed for
 at least 2 hrs.
 ↓
 filtered it, then the filtrate is ~~washed with~~
~~cold water~~ discarded and
 ↓
Residue is collected, washed with acidified water
 ↓
 Dry to obtain dark brown amorphous powder.
 ↓
 then the residue is extracted with hot alcohol,
 filtered and evaporate the alcohol and
 dry the residue
 ↓
re-crystallize in benzene } yield podophyllotoxin.

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Identification

Sample
+
50% sulphuric acid } violet-blue color.

Analysis

• Sample preparation -

1 mg of podophyllotoxin
is dissolved in 1ml of
ethanol.

Sample
+
alcohol
+
1 drop of FeCl₃ } Dark green color

• Standard sample -

podophyllotoxin

• Stationary phase → Silica gel-G

• Mobile phase - Toluene : Ethyl acetate (5:7)

• Detecting agents → spray with 50% sulphuric acid heated at 120°C for 10 min.

• Rf value → 0.39 with violet color ~~spot~~ (in day light).

- A dark brown spot appeared on TLC plate

• Uses -

- Cathartics
- purgatives
- Anti-viral agent
- Antitumor agents.

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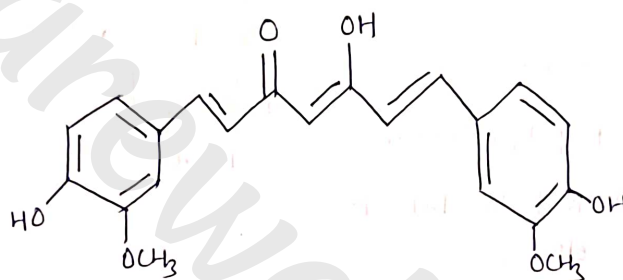
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ii) Curcumin → (curcuminoids)

It is a diaryl heptoid compounds.

Obtained from the dried rhizomes of turmeric.

belongs to family - Zingiberaceae.



• Isolation →

turmeric powder is extracted with 95% alcohol in soxhlet apparatus



extract is filter and concentrated to semisolid residue



Now, this semisolid residue is dissolved in sufficient quantity of benzene



Add 0.1% NaOH soln, shaked slowly then alkali layer is separated.



Acidify with HCl, yellow color ppt is formed

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↓
extract is concentrated with continuous stirring

↓
lumpy mass of resin will be separated out.

↓
The extract is filtered and evaporate the filtrate to get crystal of pure curcumin.

• Identification →

• Sample + acetic anhydride + conc. H_2SO_4 → violet color
observed under UV light → Red fluorescence

• Sample + H_2SO_4 → Red color.

• Analysis → (By TLC)

Sample preparation → 1mg of curcumin is dissolved in 1ml of methanol

standard sample → Curcumin

stationary phase → Silica gel-G

Mobile phase → chloroform: ethanol: Glacial acetic acid (94:5:1)

Detecting agent → observed under U.V. light at 366nm.

Rf value → 0.79

Uses →
anti-inflammatory,
anti-arthritis, antimicrobial,
antioxidant. etc.

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